AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2018

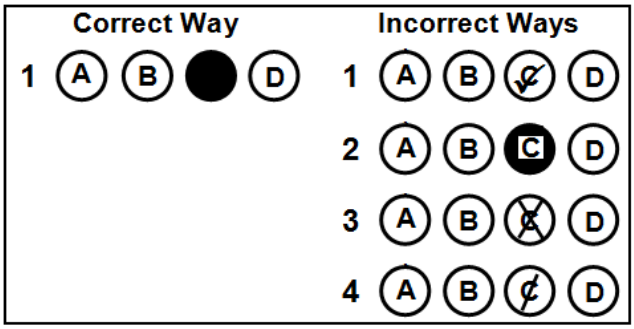
Biology Paper I

Time: 40 minutes Marks: 30

AKU-EB May 2018 for Teaching & Learning Only

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

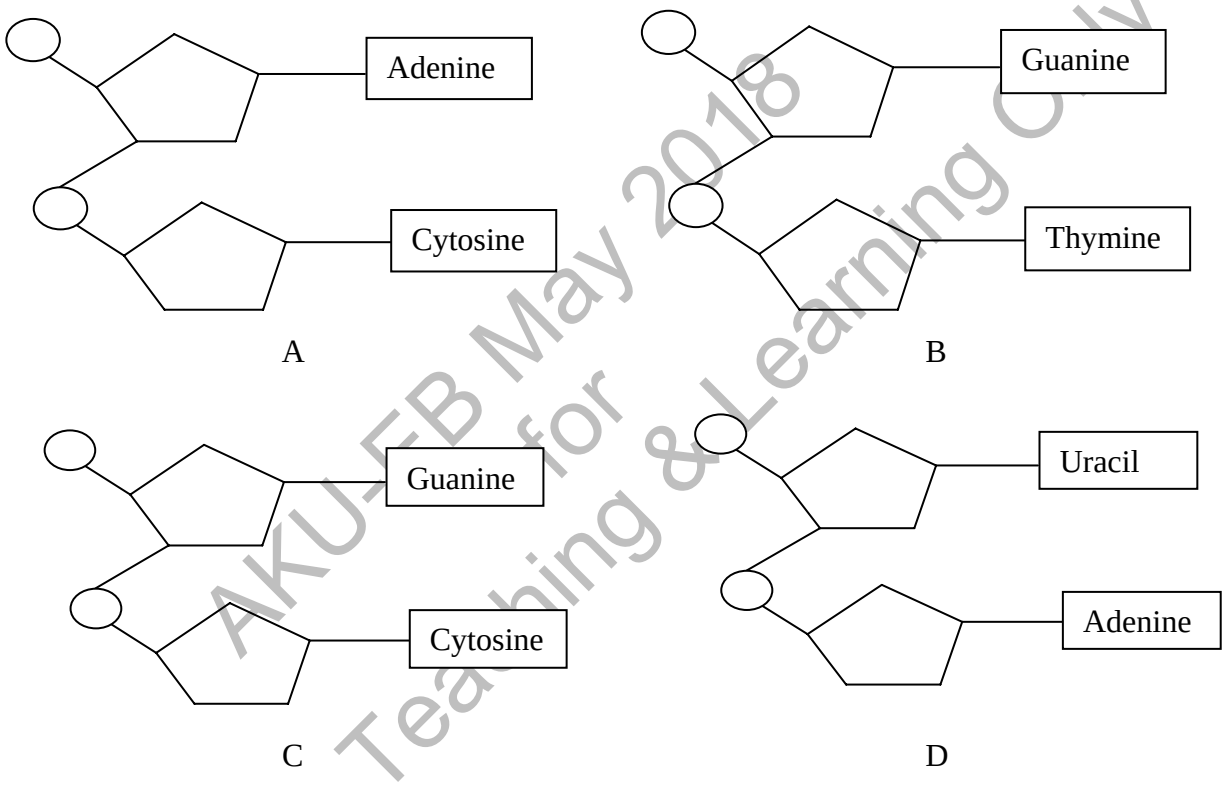
5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.

1. The jelly fish found in the Pacific Ocean have specialised stinging cells in their ectodermal layer called cnidocytes which contain a bristle like structure called cnidocil. This structure triggers cnidocytes to respond in danger situation.

   The CORRECT representation of the hierarchy of biological levels of organisation in ascending order in the given scenario is

   A. Pacific Ocean, jelly fish, ectodermal layer, cnidocytes, cnidocil.
   B. Pacific Ocean, jelly fish, ectodermal layer, cnidocil, cnidocytes.
   C. cnidocil, cnidocytes, ectodermal layer, jelly fish, Pacific Ocean.
   D. cnidocytes, cnidocil, ectodermal layer, jelly fish, Pacific Ocean.

2. All of the following illustrations represent fragments of ribonucleic acid (RNA) polynucleotide EXCEPT



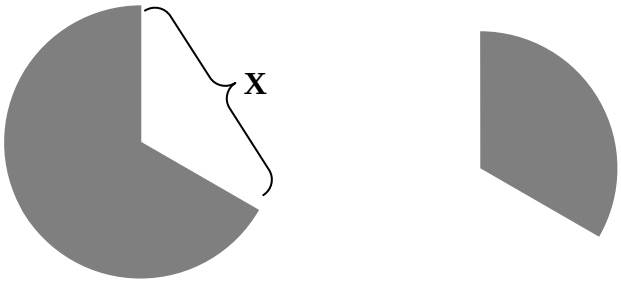
3. Zoha burns her finger by touching the side of an iron pot on the stove even though the water in the pot is still lukewarm.

   The property due to which there is a difference between the temperature of water and iron pot is

   A. density.
   B. specific heat.
   C. boiling point.
   D. heat of vaporisation.

AKU-EB May 2018 for Teaching & Learning Only

4. The given diagram shows an enzyme and a substrate.



Enzyme Substrate

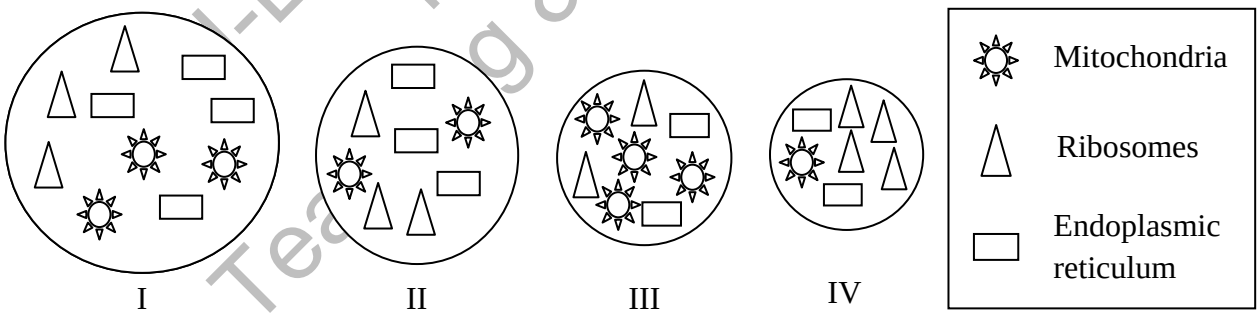
The substrate can fit at site **X** of the enzyme because enzyme and substrate have similar

A. types of amino acids.
B. chemical composition.
C. number of amino acids.
D. geometric configuration.

5. The technique which is used in the laboratory to separate charged molecules, like deoxyribo nucleic acid (DNA), according to their size is

A. centrifugation.
B. electrophoresis.
C. chromatography.
D. spectrophotometry.

6. The following diagrams represent the relative quantity of three organelles in animal cells.



Which of the given cells would have the HIGHEST energy demand for metabolic activities?

A. I
B. II
C. III
D. IV

T-EB May 2018 for Teaching & Learning Only

7. Human ovaries and testes produce steroid hormones.

   The organelle that would be abundant in human ovaries and testes to facilitate the production of steroid hormones is

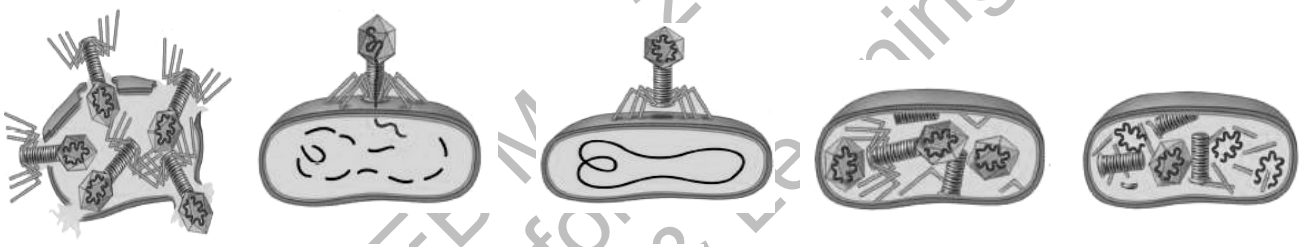
   A. lysosomes.
   B. Golgi bodies.
   C. rough endoplasmic reticulum.
   D. smooth endoplasmic reticulum.

8. Lion, tiger and cat belong to the family Felidae. The scientific name of lion, tiger and cat are *Panthera leo, Panthera tigris* and *Felis catus* respectively.

   The scientific names of lion, tiger and cat show that the cat belongs to a different

   A. class than lion and tiger.
   B. order than lion and tiger.
   C. genus than lion and tiger.
   D. phylum than lion and tiger.

9. The given diagrams illustrate different steps of lytic cycle.

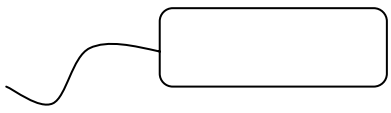


I II III IV V

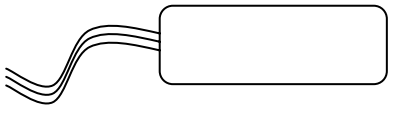
   The CORRECT sequence of their occurrence is

   A. II → III → IV → V → I
   B. II → V → III → IV → I
   C. III → II → IV → V → I
   D. III → II → V → IV → I

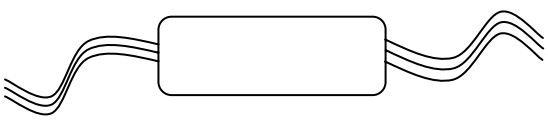
10. Which of the following diagrams represents the amphitrichous condition of bacteria?



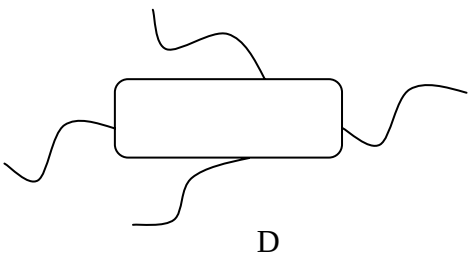
A



B



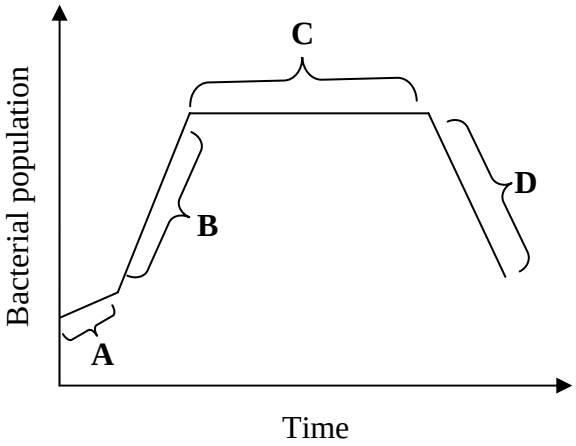
C



D

AKU-EB Material for 2018 Teaching & Learning Only

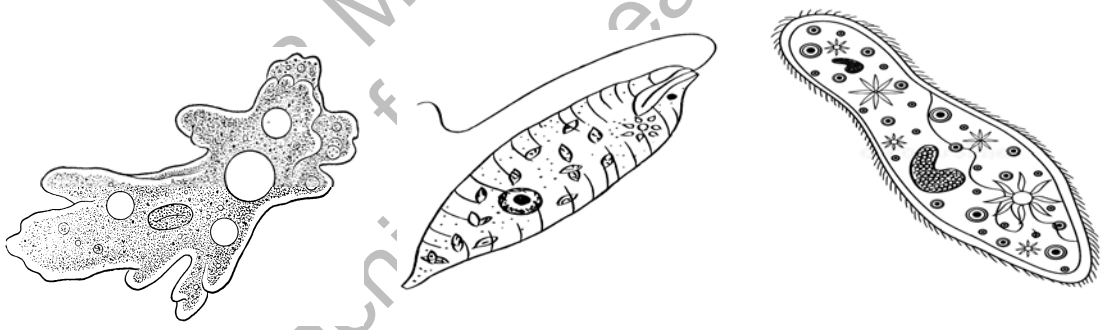
11. Which labelled phase of the given hypothetical bacterial growth curve shows the equal rate of bacterial death and multiplication?



12. Obligate anaerobes are defined as bacteria which

A. are killed by free oxygen.
B. can live with free oxygen.
C. require free nitrogen to grow.
D. can live with or without free oxygen.

13. The given illustrations show three types of protists.



The illustrated protists exhibit diversity in terms of their

A. nutrition.
B. locomotion.
C. reproduction.
D. cellular organisation.

14. In fresh water ciliates, water regulation is controlled through

A. cilia.
B. pellicle.
C. oral groove.
D. contractile vacuoles.

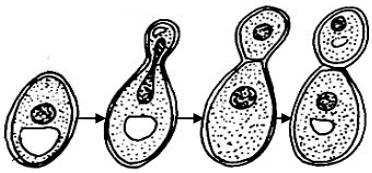
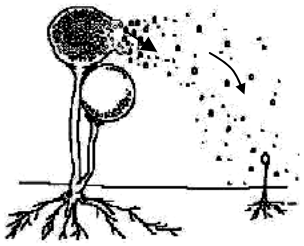
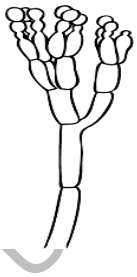
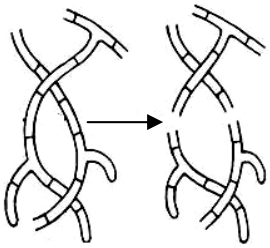
© May 2018 Teaching & Learning Only

15. The plant-like protists which possess cells covered with shells of interlocking cellulose plates are

    A. red algae.
    B. euglenoids.
    C. brown algae.
    D. dinoflagellates.

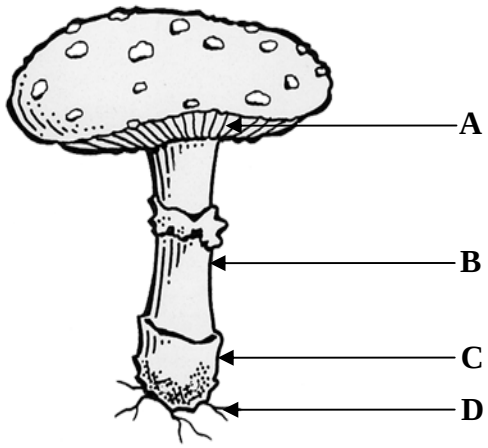
16. The given diagrams show different methods of asexual reproduction in fungi.

    Which diagram shows reproduction through conidia?

	
A	B
	
C	D

17. The given diagram illustrates a type of fungi.

    Which labelled part of the given diagram contains cells where sexual reproduction takes place?



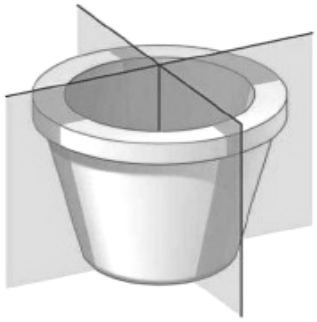
18. The given characteristics are present in a sporophyte.

- Waxy cuticle
- Meristematic tissues
- Stomata and chloroplast

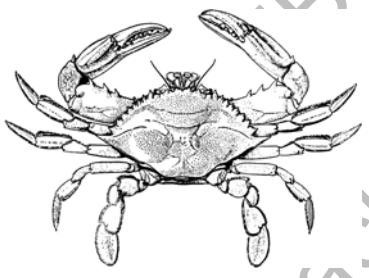
The subdivision of bryophytes that exhibits the given sporophytic characteristics is

A. Psilopsida.
B. Bryopsida.
C. Hepaticopsida.
D. Anthoceropsida.

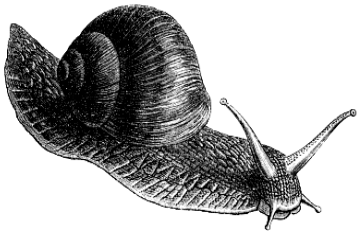
19. The given diagram represents a type of body symmetry.



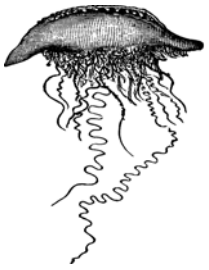
Which of the following animals has the type of body symmetry as shown in the given diagram?



A



B



C



D

20. Which of the following is CORRECT for glycolysis?

(ATP = Adenosine triphosphate, NADH = Nicotinamide adenine dinucleotide)

	Utilisation of ATP	**Production of NADH**
A	Yes	Yes
B	No	Yes
C	Yes	No
D	No	No

SEAB May 2018 Teaching and Learning Only

21. When one molecule of acetyl CoA completes the Kreb cycle it yields

A. 1 NADH, 2 $FADH_2$ and 1 ATP
B. 2 NADH, 1 $FADH_2$ and 2 ATP
C. 3 NADH, 1 $FADH_2$ and 1 ATP
D. 6 NADH, 2 $FADH_2$ and 2 ATP

22. Hina is very conscious about her body shape. She overeats and then forces herself to vomit in order to avoid gaining weight.

What condition does Hina have?

A. Botulism
B. Dyspepsia
C. Bulimia nervosa
D. Anorexia nervosa

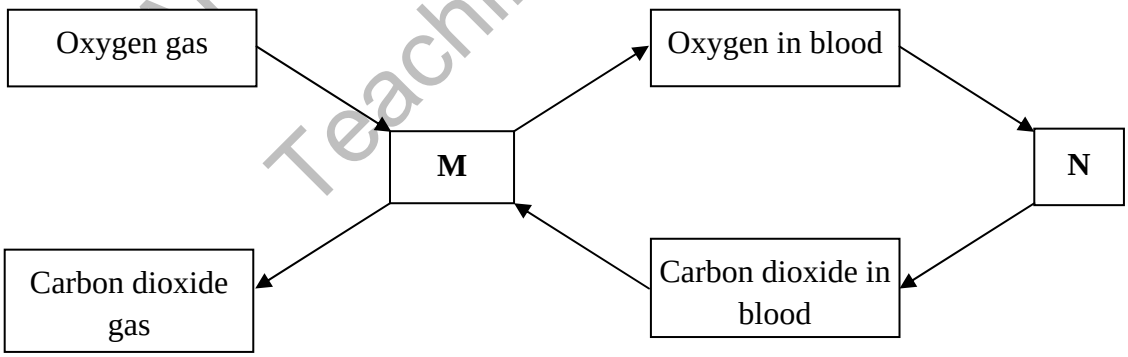
23. Plants showing stunted growth and strong chlorosis should be supplied with

A. iron.
B. calcium.
C. nitrogen.
D. magnesium.

24. The end products formed in the process of photorespiration is

A. oxygen gas and glycolate.
B. carbon dioxide gas and serine.
C. adenosine triphosphate (ATP) and ribulose biphosphate.
D. nicotinamide adenine dinucleotide phosphate (NADPH) and glycine.

25. What is represented by **M** and **N** in the given diagram?



	M	**N**
A	Alveoli	Atmosphere
B	Alveoli	Body cells
C	Body cells	Atmosphere
D	Body cells	Alveoli

AKU-EB May 2018 for Teaching & Learning Only

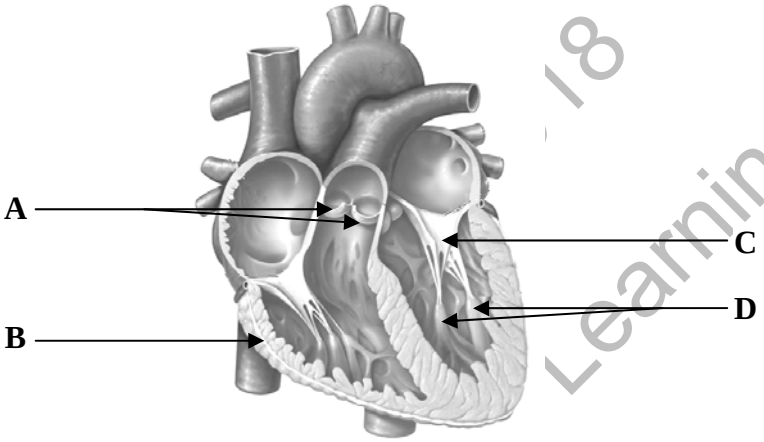
26. In fish, the surface area of the gills for gaseous exchange is increased due to the presence of

A. alveoli.
B. gill arch.
C. lamellae.
D. opercula.

27. Cells having functional similarity with lysosomes are

A. basophils.
B. neutrophils.
C. eosinophils.
D. lymphocytes.

28. Which of the labelled structures in the given diagram of the human heart indicates papillary muscles?



29. In the procedure of blood sample collection, a tube with a top containing heparin is used.

The function of heparin in blood sample collection tube is to

A. maintain blood pH.
B. prevent blood clotting.
C. prevent blood cells lysis.
D. maintain blood composition.

30. A new born baby is diagnosed with a condition in which ductus arteriosus fails to close which results in blueness of his skin. The skin of the baby appears blue because of the mixing of blood between

A. pulmonary artery and aorta.
B. left atrium and left ventricle.
C. right atrium and right ventricle.
D. pulmonary artery and pulmonary vein.

AKU-[illegible]18
Teaching & Learning Only

END OF PAPER

Please use this page for rough work

AKU-EB May 2018
for
Teaching & Learning Only

Please use this page for rough work

AKU-EB May 2018
for
Teaching & Learning Only

Please use this page for rough work

AKU-EB May 2018
for
Teaching & Learning Only